

International Islamic University Chittagong

Center for General Education (CGED)

Midterm Examination, Spring-2025

Course Code: **GEED-1201** Course Title: Basic Principles of Islam

Full Marks: 30

Time: 1 hour & 30 minutes

Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question.

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	"Islam is the guarantee of peace for the whole of the world"- explain this statement mentioning the definition and characteristics of Islam.	10	2	Create
2	(a) "Shirk is the greatest sin which destroys our 'Ibadah"- evaluate this statement explaining some types and examples of Shirk in our contemporary society. Or, (b) Distinguish between the holy Qur'an and other revealed books with proper evidence proving the authenticity of the holy Qur'an.	10	2	Evaluate & Create Analyze
3	Summarize the different stages of Akhirah. Explain the impact of belief in Akhirah on the individual and collective life of Muslims.	10	2	Remember & Create.



International Islamic University Chittagong
Department of Computer Science & Engineering
B.Sc. in CSE, Mid Term Examination, Spring 2025
Course Code: CSE-1221 Title: Computer Programming 2

Total Marks: 30 Time: 90 minutes

Answer all 3 from the following Questions. Figures in the right-hand margin indicate full marks

Answers must be done sequentially

- #
- 1 a) Describe the characteristics of OOP with examples. Marks COs DL
3 CO1 C1
- b) Write a class called 'Book' having two private member variables: 'title' and 'author'. 2 CO1 C3
 - Define a parameterized constructor to initialize them.
 - In the 'main()' function, create two objects and display their data using a member function 'display_info()'.
- c) Identify and explain the error in the following program: 2 CO1 C2
- ```

class Student {
 int a = 10;
 float g;
public:
 void display() {
 cout << a << endl; }
};

int main() {
 Student st;
 st.g = 3.75;
 st.display();
}

```

```

Or,
class Student {
 int id = 101;
public:
 float grade;
 void display(); } };
 void display() {
 cout << id << endl; }
 main() {
 Student st;
 st.grade = 4.0;
 st.display();
 }

```
- d) Explain the difference between a structure and a class in C++. Why is class preferred for real-world programming? Use appropriate code or examples. 3 CO1 C2
- or, Describe what problems associated with structured programming are solved by object-oriented programming and how?
- 2 a) What is automatic inline function? When should we use inline and when shouldn't? Give an example. 2 CO1 C2
- or, Compare and explain access specifiers: public, private, and protected.
- b) Correct the following code and explain the reason: 3 CO1 C2
- ```

#include<iostream>
using namespace std;
class Demo {
    int num;
public:
    Demo(int num) {
        this.num = num;
        cout << "Constructor: " << num << endl; }
    ~Demo() {
        cout << "Destructor: " << num << endl; }
};

int main() {
    Demo d1(10);
}
    
```



```

Demo d2(20);
{ Demo d3(30); }
cout << "Inside main function\n";
return 0; }

```

- c) Write a class named **Item** with private members: **item_id**, **item_name**, **manufacturer**, **5 CO2 C3** and **price**. Include both default and parameterized constructors, and a **set_item()** function using the **this** pointer to assign values from user input. Create an array of objects in **main()**, take input for **n** items, and display details of a specific item by matching **item_id**. Also, write a non-member function named **ApplyDiscount()** to modify the price of an **Item** and demonstrate it in **main()**.
- 3 a) Why would someone overload a function? How can a compiler identify an overloaded 3 CO2 C2 function? Describe using an example.

or, Write a program to demonstrate **constructor overloading** and explain how different constructors are invoked for different objects.

b) #include <iostream> 5 CO2 C3
using namespace std;
void display(int a = 5, int b = 10) {
 cout << "a = " << a << ", b = " << b << endl;
}
void display (double x) {
 cout << "x = " << x << endl;
}
int main() {
 display();
 display(15);
 display(15, 25);
 display(10.5);
 return 0;
}

- i) Identify and explain the **output** of the above program.
- ii) What kind of **overloading ambiguity** could arise if we add another **display(int)** function? Justify your answer.
- iii) Rewrite the above program 3b(ii) by removing ambiguity, ensuring function overloading works correctly with **default arguments**.

c) Why we use copy constructor? Write an example of copy constructor.

2 CO1 C1

```

class Person {
public:
    string name;
    void

```


[Answer all the questions; Figures in the right-hand margin indicate full marks.]

1. a) Suppose, an electronic device requires the same output polarity for any polarity of the ac input. Design a rectifier using two diodes to get the required output and derive the efficiency of a full-wave rectifier. 6 CO1 A
- b) A full-wave rectifier uses two diodes, the internal resistance of each diode may be assumed constant at $20\ \Omega$. The transformer r.m.s. secondary voltage from center tap to each end of secondary is 50 V and load resistance is $980\ \Omega$. Find: (i) mean load current, (ii) r.m.s. value of load current, (iii) rectifier efficiency. 4 CO3 A
2. a) Define transistor biasing. Derive expressions for collector current and collector-emitter voltage for an amplifier that uses voltage divider biasing method. 6 CO2 U

Or

Draw the three transistor configurations – Common Base, Common Emitter, and Common Collector configurations. For a Common Emitter configuration, establish the relation between α and β as given by: $\beta = \frac{\alpha}{1-\alpha}$

- b) A transistor uses voltage divider method of biasing. $R_1 = 50\text{k}\Omega$, $R_2 = 10\text{k}\Omega$ and $R_E = 1\text{k}\Omega$. If $V_{CC} = 12\text{V}$, find 4 CO3 A
 - (i) The value of I_C if $V_{BE} = 0.1\text{V}$.
 - (ii) The value of I_C if $V_{BE} = 0.3\text{V}$. Comment on the result.

Or

Determine the values of $V_{CE(\text{OFF})}$ and $I_{C(\text{SAT})}$ for the circuit shown in Fig. 2(b) or.

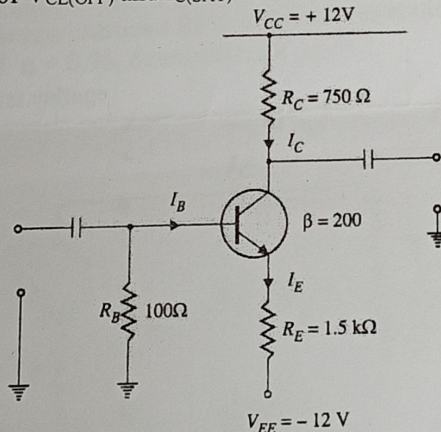


Fig. 2(b) or

3. a) Suppose, you need to design an electronic device that can perform wave shaping in such a way that it can clip out some portion of the input ac signal. Which device will you use? Define it and explain its different types with necessary figures. 7 CO2 A
- b) For the input wave to the clipping circuit shown in Fig. 3(b), find the output waveform. 3 CO3 A

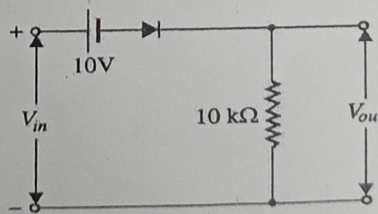
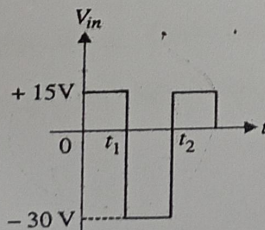


Fig. 3(b)

[Answer all the questions. Figures in the right hand margin indicate full marks.]

1. a) Define electric dipole. CLO1 R 1
 b) Obtain an expression for the electric potential and electric field due to an electric dipole. CLO2 U 6
 c) Calculate the electric potential due to a short dipole of dipole moment 4.5×10^{-25} coul-metre at a distance of 2×10^{-8} cm on a line making an angle 45° with the dipole axis from the centre of the dipole. CLO3 E 3

2. a) State Biot-Savart law. CLO1 R 1
 b) Using Biot-Savart law, calculate magnetic induction due to a current in a circular coil of wire at a point on its axis. CLO2 U 6

OR

 Define coefficient of self-induction of a coil. Deduce a mathematical expression for the self inductance of a solenoid.
 c) A circular coil has a radius of 0.15 m and number of turns of 59. Calculate the magnetic induction at a point on the axis of the coil and distance 0.25m from the centre, when a current of 0.20 A flows in it. CLO3 E 3

3. a) What is RC Circuit? CLO1 R 1
 b) Obtain an expression for the decay of charge in a capacitor connected by a D.C. battery source with a resistance in series. CLO2 U 5
 c) A $50 \mu\text{F}$ capacitor initially uncharged is connected through a 300 ohm resistor to a 12 V battery. CLO3 E 4
 (a) What is the magnitude of the final charge q_0 on the capacitor?
 (b) What is time constant of the circuit?
 (c) How long after the capacitor is connected to the battery will it be charged to $\frac{1}{2} q_0$?
 (d) How long after the capacitor is connected to the battery will it be charged to $0.90 q_0$?

OR

The distance r between the electron and the proton in the hydrogen atom is about 5.3×10^{-11} metre. What are the magnitudes of (a) the electrical force and (b) the gravitational force between these two particles?

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Mid Term Examination

Program: B. Sc. in CSE
 Course Code: **MATH-1207**
 Time: 1:30 hours



Semester: **Spring-2025**
 Course Title: **Mathematics-II**
 Total Marks: 30

- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Please answer the several parts of a question sequentially.
 (iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1:	Demonstrate knowledge of geometry and its applications in the real life contexts as well as into complex engineering problems.
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Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

- | | | Marks | CLO | DL |
|---------------|--|-------|------|----|
| 1. a) | Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2, -1)$ and inclined at an angle $\tan^{-1}\left(\frac{-4}{3}\right)$. | 5 | CLO1 | U |
| b) | Test the nature of the conic given by the equation $6x^2 + 5xy - 6y^2 - 4x + 7y + 11 = 0$. | | | |
| b) | Show that the general equation of second degree represents two parallel lines if, $\frac{a}{h} = \frac{h}{b} = \frac{g}{f}$, Also show that the distance between them is $\frac{2\sqrt{g^2-ac}}{\sqrt{a(a+b)}}$. | 5 | CLO1 | U |
| Or) | The axes being rectangular, find the equation to the pair of straight lines meeting at the origin which are perpendicular to the pair given by the equation $ax^2 + 2hxy + by^2 = 0$. | | | |
| 2. a) | A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = \frac{4}{3}$. | 5 | CLO1 | U |
| b) | Find the equation of the plane through the points $(2, 2, 1)$ and $(9, 3, 6)$ and perpendicular to the plane $2x + 6y + 6z = 9$. | 5 | CLO1 | U |
| 3. a) | Show the lines $\frac{x+3}{2} = \frac{y+5}{3} = \frac{z-7}{-3}$ and $\frac{x+1}{4} = \frac{y+1}{5} = \frac{z+1}{-1}$ are coplanar. Also find the equation of the plane containing them. | 5 | CLO1 | U |
| b) | Find the equation of the sphere for which the circle $x^2 + y^2 + z^2 + 7y - 2z + 2 = 0$, $2x + 3y + 4z = 8$ | 5 | CLO1 | U |

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science and Engineering (CSE)
Spring- 2025 (Mid Term) Total Marks-30
Course Code: CSE 1223 Course Title: Discrete Mathematics

Answer all the following questions. Figures in the right margin indicate full marks.

- | | | CO | DI |
|--|---|-----|----|
| 1. | | | |
| a) Define cardinality of set. Explain with example. | 1 | CO1 | R |
| b) Consider the set Set Y = {3, 6, 9, 12, 15}. Write the set builder notation for Set Y.
Let Set A = {1, 2, 3}, Set B = {a, b, c}. Find the Cartesian product of Set A and Set B, denoted as A × B. | 2 | CO1 | Ap |
| c) Show how bitwise operations on bit strings can be used to find these combinations of A = {a, b, c, d, e, h}, B = {b, c, d, g, h} with universal set {a, b, c, d, e, f, g, h, i, j, k}. Use bit strings to find the intersection of these sets. | 2 | CO1 | Ap |
| d) In a coding bootcamp of 40 students, it was found that: 22 students had learned Python, 18 students had learned Java, 20 students had learned C++, 9 students had learned both Python and C++, 11 students had learned both Python and Java, 7 students had learned both Java and C++, 5 students had learned all three languages. Find the number of students who had: | 5 | CO1 | Ap |
| (i) Only Java -
(ii) Java and C++ but not Python
(iii) Only one of the programming languages
(iv) None of the programming languages
(v) At least one of the three programming languages | | | |
| or, | | | |
| In a survey of 50 students at IIUC regarding their selection in three different companies — Samsung (S), Optimizly (O), and Enosis (E) — the following data was collected: 30 students were selected in Samsung, 24 students were selected in Optimizly, 22 students were selected in Enosis, 10 students were selected in both Samsung and Enosis, 18 students were selected in both Samsung and Optimizly, 8 students were selected in both Optimizly and Enosis, 6 students were selected in all three companies. | | | |
| i) Selected only in Samsung?
ii) Selected in Optimizly and Enosis but not in Samsung?
iii) selected in only one of the companies?
iv) not selected in any of the three companies? | | | |
| 2. | | | |
| a) Show that the propositions $\neg p \vee (r \rightarrow \neg q)$ and $\neg p \vee \neg q \vee \neg r$ are logically equivalent. | 2 | CO2 | A |
| b) Show that $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology. | 2 | CO2 | A |
| c) Consider the following propositions: p: "The program runs without errors.", q: "The developer submits the code.", r: "The code passes all test cases." Translate the following propositions into English statements based on the given propositions:
i) $p \wedge q$ (Conjunction) ii) $\neg r$ (Negation) iii) $p \rightarrow r$ (Implication) and iv) $q \leftrightarrow r$ (Biconditional) | 3 | CO2 | A |
| d) Consider the following predicates: | 3 | CO2 | |

- $D(x)$: "Programming language x is dynamically typed."
- $E(x)$: "Programming language x is easy to learn."
- $T(x, y)$: "Programming language x is used to teach y ."

Let the domain be the set of all programming languages. Translate the following English statements into logical expressions using quantifiers and the given predicates:

- "All dynamically typed languages are easy to learn."
- "There exists a language that is easy to learn and is used to teach every other language."
- "If a language is dynamically typed, then it is used to teach some language that is easy to learn."

$D(x)$

$T(x, y)$

$E(y)$

3.

- Let $f: A \rightarrow B$ be a function where: $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$. The function f is defined as: $f(1)=a$, $f(2)=b$, $f(3)=c$, $f(4)=b$. Is the function f **one-to-one** and **onto**? Justify your answer. 1 CO1
 - List the first five terms of each of the following sequences: 2 CO2
 - A sequence where each term is the sum of the previous term and the term's index multiplied by 2.
 - A sequence starting with 4 and obtaining each term by multiplying the previous term by 2 and then subtracting 1.
 - Let g be the function from the set $\{1, 2, 3\}$ to itself such that: $g(1) = 2$, $g(2) = 3$, $g(3) = 1$. Let f be the function from the set $\{1, 2, 3\}$ to the set $\{10, 20, 30\}$ such that: $f(1) = 30$, $f(2) = 20$, $f(3) = 10$. What is the composition $f \circ g$? Is $g \circ f$ defined? 2 CO1
 - Let the set $P = \{\text{Python, Java, C++}\}$ represent programming languages. Each relation below is defined on set P . For each relation, determine whether it is: **i) Reflexive**, **ii) Symmetric**, **iii) Antisymmetric**, **iv) Asymmetric**, **v) Transitive** 5 CO1
 - $R_1 = \{(\text{Python, Python}), (\text{Python, Java}), (\text{Java, Python}), (\text{Java, Java}), (\text{C++}, \text{C++})\}$
 - $R_2 = \{(\text{Python, Java}), (\text{Java, C++}), (\text{C++}, \text{Python})\}$
 - $R_3 = \{(\text{Python, Python}), (\text{Java, Java}), (\text{C++}, \text{C++}), (\text{Python, Java}), (\text{Java, C++}), (\text{Python, C++})\}$
 - $R_4 = \{(\text{Python, Python}), (\text{Java, Java})\}$
- or, For each of these relations on the set $\{1, 2, 3, 4\}$, decide whether it is reflexive, whether it is symmetric, whether it is antisymmetric, and whether it is transitive.
- $R_1 = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 4), (4, 1), (4, 4)\}$,
 - $R_2 = \{(1, 1), (1, 2), (1, 4), (2, 1), (2, 2), (3, 3), (4, 1), (4, 4)\}$,
 - $R_3 = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3)\}$,
 - $R_4 = \{(3, 4), (2, 2)\}$.